Greenhouse self-sufficiency model: implemented scenarios

EPFL x Voltiris master thesis *A Path to Greenhouse Self-Sufficiency: Heat and CO_2* · model status 12 October 2026 (Baseline and Scenario 1) · all numbers come from the synthetic dummy dataset (1 ha, Swiss carbon levy) and placeholder parameters: they illustrate the model, not a real site.

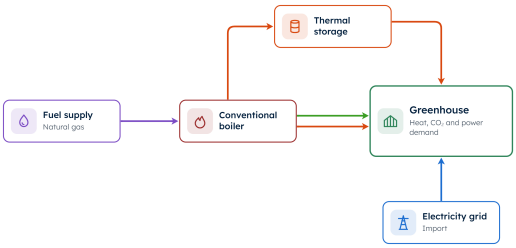
1. Common framework

Each scenario is a linear program built with PyPSA and solved with HiGHS over one year at hourly resolution (8,760 h, perfect foresight). The system is assembled from reusable **blocks** (boiler, CHP, storage, grid, ...) connected to five buses: gas, electricity, heat (supply temperature), storage interior and CO_2. The objective is the yearly cost: fuel, carbon levy, electricity purchase minus export revenue, variable O&M and penalties for unserved demand; annualised capex enters only when a size is optimised, and is always reported.

Inputs (Excel, hourly or daily)	Main parameters (Params)	Outputs (every scenario, same KPI rows)
Heat, electricity and CO_2 demand; gas price (LHV); electricity import price; grid emission factor (optional); **export price** (S1)	Efficiencies, capacities (fixed or optimised), capex, lifetimes, O&M, storage losses and limits, country (carbon levy, grid factor), discount rate, 20-year horizon	Hourly flows; fuel use; grid import/export; heat dumped, CO_2 vented; storage use; self-sufficiency; total annualised cost; emissions; storage value; payback, NPV, IRR vs a reference

Constraints common to all scenarios. (1) Hourly balance at every bus. (2) Fixed demands, met every hour; unserved heat/CO_2 is allowed only as a slack penalised at 1,000 EUR/MWh and 5,000 EUR/t (zero in all results). (3) Converters with constant efficiencies and capacity limits, no minimum load. (4) Thermal storage: energy balance with charge/discharge efficiency (0.98) and standing loss (0.1 %/h), same level at the start and end of the year, power <= capacity / 4 h, 10 % reserve. (5) Heat dump and CO_2 vent are free and unlimited: they are **assumptions** (not drawn in the scenario figure).

2. Baseline: natural gas boiler + thermal storage + grid import



System. The boiler is the only source of heat and CO_2 (flue gas). The storage buffers boiler heat; all electricity is imported. Sizes: boiler 1.2 x peak heat (1.31 MW), storage 6 h of peak heat (6.5 MWh).

Main constraints. Heat = 0.92 x gas and CO_2 = 0.202 t/MWh x gas, so heat and CO_2 are produced in a fixed ratio (about 0.22 t CO_2 per MWh of heat); boiler output <= capacity and may change by at most 50 % of capacity per hour; storage limits as above; grid import = electricity demand.

Results and limiting constraints (dummy data)

Output	Value	Constraint check	Binding?
Gas use / grid import	3,582 / 363 MWh/y	Boiler at full capacity	4 h/y: no
Heat dumped	265 MWh/y (8 % of boiler heat)	Heat/CO_2 coupling (dump active)	**1,058 h/y: yes**
CO_2 vented	484 t/y (CO2 demand 240 t/y)	CO_2 vent active	**4,490 h/y: yes**
Storage discharged	279 MWh/y, 43 cycles	Storage full / at reserve / at power limit	4 / 3,454 / 0 h/y
Total annualised cost	317.1 kEUR/y	Ramp limit reached	5 h/y: no
Total CO_2 emissions	760 t/y	Marginal value of CO_2 in dosing hours	135 EUR/t

Reading. No capacity limits the Baseline: the boiler and the storage almost never reach their limits. The **limiting constraint is the fixed heat/CO_2 ratio of the boiler**. In summer gas is burnt for CO_2 and heat is dumped; in winter heat is needed at night and CO_2 is vented. CO_2 therefore has a high marginal cost in dosing hours. The storage removes part of the conflict by shifting CO_2-driven daytime heat to the night: it saves 19.9 kEUR/y of operating cost for a 65 kEUR investment (simple payback 3.4 y, IRR 29 %, NPV 180 kEUR). It sits at its reserve most of the time: it is limited by the heat surplus available, not by its size.

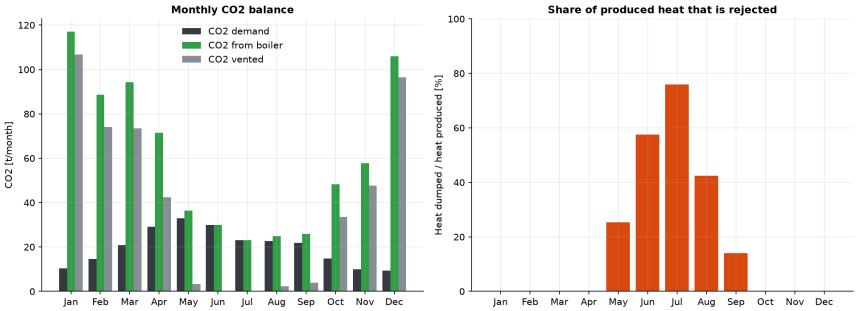
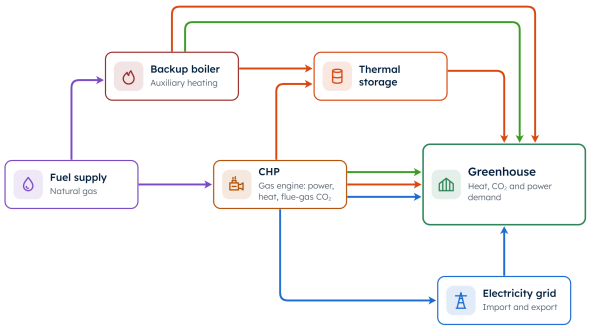


Figure 1. Baseline, dummy data: monthly CO_2 balance (left) and share of boiler heat that is dumped (right). Vented CO_2 in winter and dumped heat in summer are the two faces of the fixed heat/CO_2 ratio.

How the Baseline can be upgraded.

- **Decouple CO_2 from heat**: a CO_2 buffer or liquid CO_2 (as in the S2/S3 variants) would remove most of the dumped heat; the marginal CO_2 value above gives the price such a supply must beat.
- **Realistic CO_2 demand**: link dosing to ventilation (open vents waste CO_2) and allow partial dosing at a yield cost instead of a fixed demand.
- **Boiler operation**: minimum load and start-up costs (mixed-integer option), part-load efficiency, condensing efficiency with return temperature.
- **Storage and data**: optimised storage size (OPTIMISE_TES) with a stratified model; real hourly data per country, sourced placeholders and several weather years to test the boiler margin.

3. Scenario 1: gas CHP + backup boiler + thermal storage + grid import/export



System. A gas engine CHP produces electricity, heat and CO_2 (exhaust). The backup boiler (same block, full backup size) covers what the CHP cannot. Both feed the storage. Surplus electricity is **exported and counted as revenue** at the hourly export price (new input column); the grid covers any deficit.

Main constraints. CHP electricity = 0.42 x gas, heat = 0.45 x gas, CO_2 = 0.202 t/MWh x gas (fixed ratios, back-pressure); CHP size 0.51 MW_{el} (thermal capacity = 50 % of peak heat); electricity balance: import + CHP = demand + export. **CHP priority**: a dispatch weight of 200 EUR/MWh on boiler heat and grid import (never reported as a cost) makes the CHP the first choice; the order is CHP < boiler and import < unserved heat.

Results and limiting constraints (dummy data)

Output	Baseline	S1 with CHP priority	S1 cost-optimal (no priority)
CHP full-load hours / hours on / starts	–	5,222 h / 8,760 h / 1	623 h / 5,085 h / 378
CHP at full capacity	–	**4,192 h/y**	4 h/y
Boiler running	7,186 h/y	1,554 h/y	5,541 h/y
Gas use	3,582 MWh/y	6,627 MWh/y	3,740 MWh/y
Grid import / export	363 / 0 MWh/y	0 / 2,290 MWh/y	147 / 101 MWh/y
Export revenue	0	145.9 kEUR/y	5.5 kEUR/y
Electricity self-sufficiency	0 %	100 %	59 %
Heat dumped / CO_2 vented	265 MWh / 484 t	85 MWh / 1,099 t	53 MWh / 516 t
Total annualised cost	317.1 kEUR/y	447.7 kEUR/y	357.4 kEUR/y
Total CO_2 (export credit not subtracted)	760 t/y	1,339 t/y	770 t/y
NPV vs Baseline (20 y, 5 %) / payback	reference	-1,641 kEUR / never	-515 kEUR / 121 y (simple)

Reading. Two constraints limit S1. (1) **CHP capacity** binds for 4,192 h: in winter the backup boiler still runs 1,554 h. (2) The **heat/power/CO_2 coupling of the CHP combined with the priority rule**: running the CHP for heat forces 2,290 MWh/y of export at an average 64 EUR/MWh, while CHP electricity costs about 99 EUR/MWh after crediting its heat (gas + levy + O&M). The rule costs 90 kEUR/y compared with cost-optimal dispatch, and emissions rise by 579 t/y because exported electricity is not credited. Without the priority the CHP runs only 623 full-load hours with 378 starts, an artefact of the LP (no minimum load or start cost).

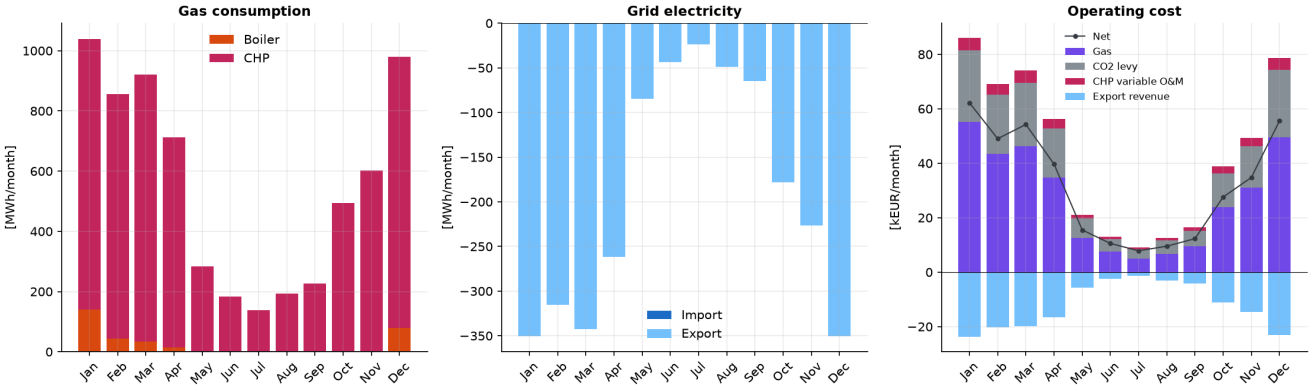


Figure 2. Scenario 1 with CHP priority, dummy data: monthly gas use per unit, grid import (none) and export, and operating cost with export revenue as a negative bar.

How Scenario 1 can be upgraded.

- **Unit commitment**: minimum load (~50 %), start-up costs and minimum run times for the CHP (mixed-integer option), so that on/off cycling and part-load operation are realistic; part-load efficiency curves.
- **CHP sizing and operating mode**: optimise the CHP size (OPTIMISE_CHP) and compare heat-led, power-led and cost-optimal dispatch; replace the soft priority by a must-run share if a strict rule is needed.
- **Electricity value**: real hourly spot prices for export, grid fees on export, capacity or flexibility revenues, and an export limit from the real connection. The CHP value comes from high-price hours, which a flat dummy price misses.
- **Electricity demand**: add the greenhouse lighting load (assimilation lighting), which raises on-site use of CHP electricity, the main lever on S1 economics.
- **CO_2 and emissions**: exhaust cleaning efficiency (usable CO_2 share below 1), and a decision on whether exported electricity is credited with avoided grid emissions (now reported separately, not subtracted).

Financial analysis of S1 against the Baseline

The additional equipment is the CHP (457 kEUR; boiler and storage are identical in both scenarios). With the priority rule the yearly cash saving is negative (-87 kEUR/y), so there is no payback and no IRR; the NPV over 20 years (CHP replaced after 15 years, salvage value at the end) is -1,641 kEUR. Holding the dispatch fixed, the export price would have to rise by about 57 EUR/MWh (to about 121 EUR/MWh) for the CHP to break even. The NPV is most sensitive to the export price (-2,004 to -1,277 kEUR for -20/+20 %) and the gas price (-2,017 to -1,265 kEUR for +20/-20 %). These signs are robust; the magnitudes depend on the dummy prices and the Swiss carbon levy (125 EUR/t).

Investment appraisal (first-order sensitivity: dispatch not re-optimised)	Additional investment	Net yearly saving	Simple / discounted payback	NPV (20 y, 5 %)	IRR
Baseline: storage vs no storage	65 kEUR	19.3 kEUR/y	3.4 / 3.8 y	180 kEUR	29 %
S1 (CHP priority) vs Baseline	457 kEUR	-86.5 kEUR/y	never	-1,641 kEUR	n/a
S1 cost-optimal vs Baseline	457 kEUR	3.8 kEUR/y	121 y / never	-515 kEUR	n/a

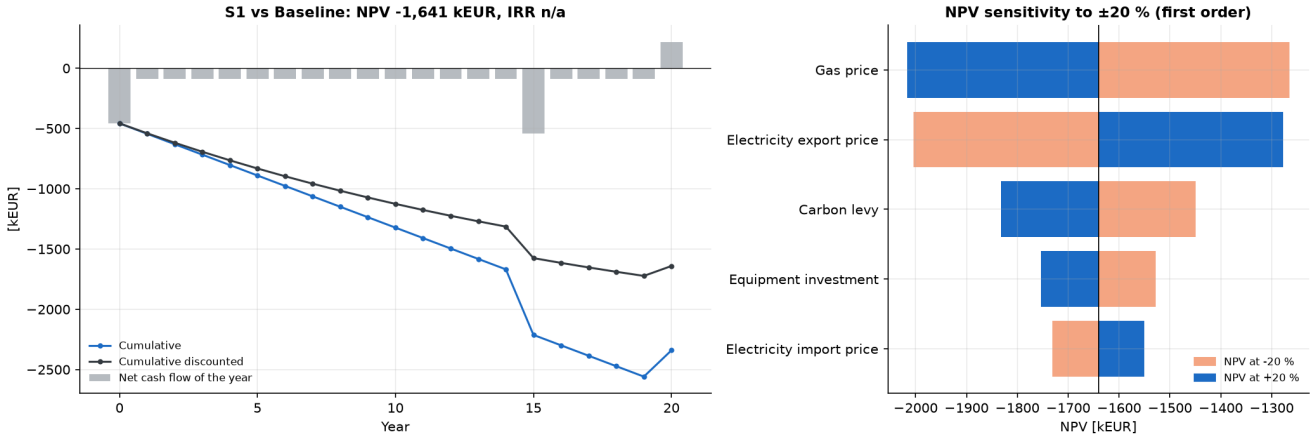


Figure 3. S1 vs Baseline: yearly and cumulative cash flows (CHP replacement in year 15, salvage value in year 20) and first-order NPV sensitivity.

4. Summary: limiting constraints and upgrade path

Limiting constraint	Baseline (evidence)	Scenario 1 (evidence)	Upgrade
Fixed heat/CO_2 ratio of gas units	dump 265 MWh, vent 484 t	vent 1,099 t (CHP runs for heat)	CO_2 buffer or liquid CO_2; dosing linked to ventilation
Unit capacity	boiler at capacity 4 h: not binding	CHP at capacity 4,192 h; boiler on 1,554 h	optimised sizes; several weather years
Value of electricity (heat/power coupling)	-	export at 64 EUR/MWh vs ~99 EUR/MWh CHP cost	spot prices, lighting load, export fees and limits
Dispatch rule	-	priority costs 90 kEUR/y	must-run share or cost-optimal dispatch
LP simplification (no minimum load)	ramp limit reached 5 h	cost-optimal CHP: 378 starts/y	mixed-integer unit commitment
Storage	at reserve 3,454 h, full 4 h	at reserve 1,444 h, full 29 h	size optimisation, stratified model

5. Cross-cutting limits and next steps

- **Data**: every result above uses synthetic data and placeholder parameters; real hourly series per country (CH, FR, NL) and sourced efficiencies and costs are needed before drawing conclusions.
- **Method**: perfect foresight over one year and constant prices over 20 years (no escalation, no financing cost, no subsidies); a multi-year or price-scenario analysis would test the robustness of the payback results.
- **Heat and CO_2 self-sufficiency** are 0 % in both scenarios (all heat and CO_2 come from gas); electricity self-sufficiency reaches 100 % with the CHP but on fossil fuel. The non-fossil scenarios S2-S4 (Voltiris PV, heat pump, battery, data-center heat, SOFC on ethanol) are the next blocks to build, reusing the same framework, KPI table and financial analysis against the Baseline.
- **Open decisions**: CO_2 accounting for ethanol (S4) and the backup heat source for the variants without a boiler.

Code, block documentation and assumptions: `src/ghmodel/`, `docs/blocks/`, `README.md` (Main assumptions). Notebooks: `notebooks/01_baseline.ipynb`, `notebooks/02_s1_chp.ipynb`.